

Decision memo

For the marketing lead. Austrian media mix and budget optimizer (AMBO), demonstration advertiser.

1. The decision this memo supports

Approve a same-budget change to the weekly media mix: more TV and Facebook, less Out-of-home, and no change to Print or Search.

2. Recommendation

Make the change. Media contribution is expected to rise by 15.1 percent, and at the 10th percentile the gain is still 8.8 percent.

3. Evidence

- Out-of-home weekly spend falls 20 percent in the recommended mix (`reports/layer_d/reallocation_table.csv`).
- The chance the change loses money is 0.0 percent (`reports/layer_d/decision.json`).
- On a synthetic advertiser with known truth, the platform report credited Paid Search with 45 percent more revenue than it really added (`reports/layer_p/attribution_gap.csv`).

4. Chart

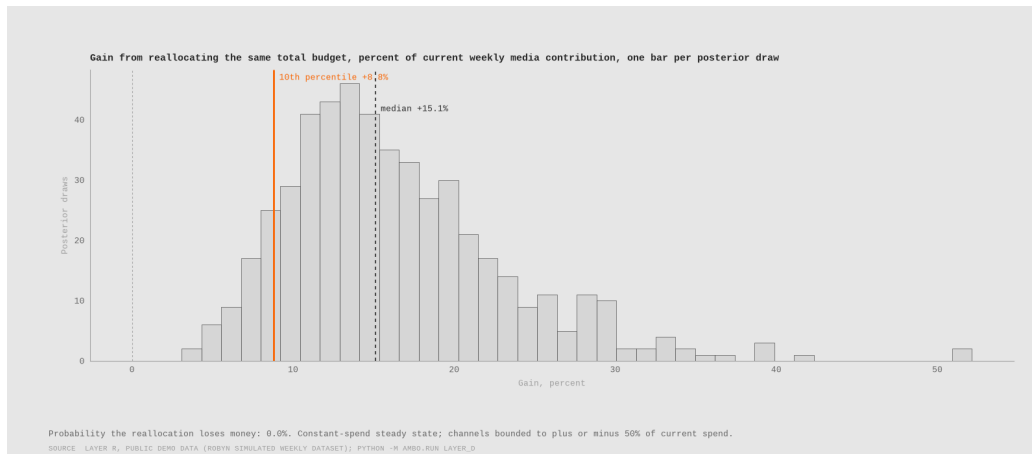


Figure 1: Gain from reallocating the same total budget, as a percent of current weekly media contribution

Main results chart: `reports/layer_d/reallocation_gain.png`. The dashed line is the 15.1 percent median; the solid line is the 8.8 percent tenth percentile. The mass of the chart sits above zero.

5. What we do not know

- The demonstration data is a public simulated dataset in unnamed money units. Shares and percent gains are usable; the cash amounts are not euros.
- No live advertiser, and no Austrian client file, has been fitted. Competitors may change spend in response.
- The recommended weeks assume the fitted curves still hold at the new spend levels; that has not been tested with a lift experiment.

6. Next step and its cost

Apply this mix on the demonstration budget until a live euro file exists. Extra media cost is 0; the weekly total stays 69851 in the dataset's units (`reports/layer_d/decision.json`).¹

Numbers in this memo

- 15.1 percent — `reports/layer_d/decision.json` (same-budget gain as a percent of current contribution, 15.105...), also `reports/readme_values.json`
- 8.8 percent — `reports/layer_d/decision.json` (10th percentile, 8.800...), also `reports/readme_values.json`
- 20 percent — `reports/layer_d/reallocation_table.csv` (Out-of-home change −19.655...), rounded as in `reports/readme_values.json`
- 0.0 percent — `reports/layer_d/decision.json` (probability the gain is negative)
- 45 percent — `reports/layer_p/attribution_gap.csv` (Paid Search platform figure over truth, 45.131...)
- 0 extra and 69851 weekly total — `reports/layer_d/decision.json` (recommended total equals current total, 69851.435...)

¹Estimates come from a Bayesian media mix model.